

Molly Beach

SOFTWARE ENGINEER



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EXECUTIVE PROFILE

Infrastructure and protocol leader; Associate Director of Blockchain Engineering at DTCC Digital Assets (institutional tokenization, Factory platform, Stellar/Soroban). Protocol engineer specializing in consensus systems, distributed execution, and cryptographic validation architectures. Proven record designing and leading complex platform systems that operate across multi-language execution environments, distributed validators, and adversarial conditions. Recognized for architectural ownership, system-level decision making, and building infrastructure that scales across networks, teams, and production environments.

Core strengths: Consensus + validator architecture · Cryptographic verification systems and signatures · Distributed infrastructure design · Protocol-level smart contracts · Incentive + slashing mechanism design · Adversarial modeling + attack resistance · Cross-chain execution systems · Fault tolerant system design

EXPERIENCE



DTCC DIGITAL ASSETS

APR 2026 - PRESENT, REMOTE

ASSOCIATE DIRECTOR OF BLOCKCHAIN ENGINEERING | RUST, SOROBAN, SOLIDITY, MULTI-CHAIN

Associate Director leading blockchain engineering for DTCC's institutional tokenization platform—extending production Factory orchestration (Ethereum, Hyperledger Besu, Canton) to Stellar/Soroban and a multi-chain roadmap including Solana. Senior technical and people leadership across smart contracts, enterprise SDLC, and regulated market infrastructure.

Factory Platform & Multi-Chain Architecture

- Set technical direction for Soroban (Rust) implementation of DTCC's Factory smart-contract orchestration—factories, templates, strategies, and registries for tokenized issuance, permissions, compliance, and custody integration
- Lead replication of mature Solidity/EVM contract systems into Soroban while preserving compliance, governance, auditability, and operational controls at institutional scale
- Partner across Solidity, front-end, indexing, design, and business analysis in scrum delivery for chain-agnostic tokenization (pilots: treasuries, ETFs; SEC no-action institutional inventory on-chain)

Enterprise Delivery & People Leadership

- Own enterprise testing and security workflows: fuzz, invariant, and mutation testing, static analysis, coverage, audit readiness, monitoring/observability, and Tier 2 / permit-to-operate expectations
- Integrate Stellar into LedgerScan via Firehose and Substreams; GraphQL-style indexed data for internal systems, analytics, and auditable operational views
- Hire and grow blockchain engineering talent; balance hands-on architecture with coaching, process, and cross-functional alignment in a remote global team

Rust · Soroban · Solidity · Stellar · Solana · Factory architecture · Firehose · Substreams · GraphQL · React · TypeScript · CI/CD · Enterprise SDLC · Tokenization



BLOCKMEDIA

MAY 2025 - MAR 2026, REMOTE

PRINCIPAL PROTOCOL & INFRASTRUCTURE ENGINEER | GO, RUST, PYTHON

Lead architect for a Layer 1 EVM-compatible blockchain integrating federated machine learning into validator consensus, supporting an ecosystem with 34K+ token holders and ~\$20M+ raised, requiring production-grade consensus reliability and adversarial-resilient protocol design.

Architectural Leadership and System Ownership

- Architected validator consensus extension enabling off-chain AI model execution verification and on-chain settlement
- Designed transaction schema + execution state machine for training, inference, and reward distribution flows
- Defined slashing + incentive model for malicious or low-quality validator submissions
- Implemented model validation scoring pipeline (perplexity + accuracy metrics) used for consensus-level reward decisions
- Designed adversarial-resistant validation pipeline preventing poisoned model updates and replay attacks
- Set architectural standards for deterministic validation and cross-node consensus guarantees

Platform Scale and Systems Impact

- Designed multi language validator execution architecture coordinating Go, Rust, Python, & Solidity layers
- Built monitoring and telemetry systems measuring validator participation and consensus health
- Implemented asynchronous execution endpoints used by distributed validator nodes
- Integrated IPFS checkpoint storage ensuring reproducible validation across nodes

SKILLS

Programming Languages

> TypeScript
> JavaScript
> Solidity
> Python
> Swift
> Rust
> Java
> Golang
> Scala
> C++
> R
> JSON
> Ruby
> HTML
> CSS
> SCSS

Frontend Development

> React
> Node.js
> Vue
> Next.js
> Angular
> D3.js
> Tailwind
> Bootstrap
> Material UI
> State Management
> Redux
> Zustand

Backend Development & API Integration

> Node.js
> Express.js
> Django
> Flask
> GraphQL
> REST APIs
> Web Sockets
> OAuth
> JWT
> Firebase Auth

Databases & Data Management

> PostgreSQL
> SQL
> MySQL
> MS SQL Server
> Firestore
> MongoDB
> Redis
> Docker
> Pandas
> NumPy
> TensorFlow
> Matplotlib

Cloud, DevOps & Infrastructure

> AWS
> Microsoft Azure

Cryptographic and Consensus Engineering

- Implemented validator verification using Ed25519 and BLS aggregation
- Designed deterministic model verification and byte range hashing for cross node agreement
- Architected reward contract system governing distribution, accounting, and DAO authority
- Defined blob transaction flow carrying model diffs and proofs across network participants

Go · Rust · Python · Solidity · PyTorch · Solidity · Docker · Hardhat · Federated Learning · Cryptography · EVM Forks · Protocol Design · AI



BRKT

JAN 2025 - APRIL 2025, REMOTE

SENIOR SOFTWARE ENGINEER | PROTOCOL + INFRASTRUCTURE FOCUS

Built core infrastructure for decentralized prediction and competition markets designed to support large-scale bracket tournaments and real-time peer-to-peer wagering, within the \$30B+ online gambling and prediction market ecosystem.

Protocol Architecture and Execution

- Architected settlement engine designed high-frequency wager execution under real-time odds changes
- Designed event-driven wager execution engine handling real-time odds updates and state transitions
- Architected low-latency settlement APIs supporting concurrent transaction flows
- Defined + implemented ranking + payout engine processing high-frequency user events

Scalability and Market Positioning

- Designed systems positioned to support large bracket competitions (up to 256 participants) for major annual events (e.g., March Madness, NFL Finals) by enabling fast, trustless market interactions
- Developed infrastructure with future-focused mobile engagement pathways in anticipation of broad user adoption, aligning with industry growth trends in wagering and prediction markets

Security / Verification Work

- Optimized system for sub-second event propagation across distributed clients
- Designed infrastructure supporting high-concurrency wager placement without state conflicts

Performance and Reliability

- Optimized system for sub-second event propagation across distributed clients
- Designed infrastructure supporting high-concurrency wager placement without state conflicts
- Engineered ranking and payout engine to efficiently process high-frequency state transitions and outcomes

TypeScript · Rust · React Native · Node.js · GraphQL · gRPC · PostgreSQL · Redis · WebSockets · Zustand · Expo · Reanimated · Docker · Kubernetes · Nginx · Firebase · OAuth · JWT · CI/CD



PARADES

JAN 2024 - DEC 2024, REMOTE

SENIOR FULL-STACK SOFTWARE ENGINEER | DISTRIBUTED SYSTEMS

Focused on backend systems, distributed data platforms, and production infrastructure reliability.

- Built distributed data services using Scala + Akka for fault-tolerant processing
- Designed APIs handling high-volume transactional workloads
- Deployed microservices across AWS + GCP environments with production monitoring
- Conducted performance tuning across memory scheduling and compute pipelines

Swift · Scala · iOS Development · Functional Programming · Akka · Microservices · REST APIs · PostgreSQL · MongoDB · AWS · GCP · Performance Optimization · UI/UX Design · Agile Development · Cross-functional Collaboration



ACCENTURE

OCT 2022 - DEC 2023, REMOTE

SOFTWARE ENGINEER FULL-STACK | METAVERSE, BLOCKCHAIN INFRASTRUCTURE

Led enterprise blockchain systems, cross-chain verification infrastructure, and cryptographic validation services.

- Architected consortium blockchain systems using Hyperledger Fabric and Besu
- Implemented validator-style verification logic in Go + Rust services
- Optimized runtime performance under load via concurrency + memory tuning
- Designed cross-chain interoperability architecture using Polygon Edge + Kaleido

Impact & Ownership

- Developed credential validation automation for 96 services detecting config drift and redeploying compromised nodes
 - Built event listeners monitoring blockchain state and triggering automated enforcement logic
- Rust · Go · Solidity · TS · Python · JS · React · Ethereum · Polygon · Hyperledger Fabric · Besu · GoQuorum · Hardhat · Ethers · Account Abstraction · AWS · Azure · Docker · Serverless Swagger · Nanostores · Tailwind · 3JS · Unity · Blender · Power Automate · Canvas API · Metaverse · Securiti



PARADES

SEP 2021 - SEP 2022, REMOTE

SOFTWARE ENGINEER FULL-STACK | DISTRIBUTED SYSTEMS + SMART CONTRACTS

- Designed modular smart contract infrastructure deployed across EVM networks
- Architected concurrent Rust validation modules for high-throughput data pipelines

- > Vercel
- > Netlify
- > Heroku
- > Cloudflare
- > GCP
- > Docker
- > Kubernetes
- > Terraform
- > GitHub Actions
- > Jenkins
- > GitLab CI/CD
- > Firebase Functions
- > AWS Lambda

Blockchain & Web3

- > Solidity
- > Ethereum
- > Stellar
- > Soroban
- > Aptos
- > Near
- > Arbitrum
- > Solana
- > Hardhat
- > Truffle
- > Remix
- > Ganache
- > Foundry
- > IPFS
- > Moralis
- > Chainlink
- > Arweave

Smart Contract & Security

- > Zero-knowledge Proofs
- > Account Abstraction
- > Reentrancy Prevention
- > Delegatecall Safety
- > ECDSA Signature Verification
- > Access Control Security

Message Queues & Event Driven Architecture

- > Apache Kafka
- > RabbitMQ
- > ZeroMQ
- > Redis Streams
- > NATS
- > Event Sourcing
- > CQRS
- > Webhooks
- > Pub/Sub Architecture
- > Google Pub/Sub
- > AWS SNS/SQS

System Design & Scalability

- > Distributed Systems
- > Microservices
- > Event Driven Architecture
- > Caching & Load Balancing
- > Redis
- > Memcached
- > Nginx
- > Cloudflare

Testing & QA

- > Jest
- > Mocha/Chai

- Defined + implemented automated contract testing + CI pipelines ensuring deterministic releases
- Delivered production monitoring + infrastructure observability

Impact & Ownership

- Developed credential validation automation for 96 services detecting config drift and redeploying compromised nodes
- Built event listeners monitoring blockchain state and triggering automated enforcement logic

Solidity · TypeScript · JavaScript · Hardhat · Foundry · Node.js · React · Ethers.js · Web3 · IPFS · GraphQL · Docker · GitHub Actions Python · Rust · Flask · PostgreSQL · Celery · Redis · Docker



TREASURE

JAN 2020 - AUG 2021, REMOTE

SOFTWARE ENGINEER FULL-STACK | BLOCKCHAIN SYSTEMS

- Designed modular smart contract infrastructure deployed across EVM networks
- Architected concurrent Rust validation modules for high-throughput data pipelines
- Implemented automated contract testing + CI pipelines ensuring deterministic releases
- Delivered production monitoring + infrastructure observability

Solidity · Rust · TypeScript · Python · JavaScript · Ethers.js · Web3 · Hardhat · Foundry · IPFS · Node.js · Flask · GraphQL · React · PostgreSQL · Redis · Celery · Docker · GitHub Actions

PROJECTS



CRYPTO GENE

MAR 2020 - DEC 2020

Developed a DApp/Web3JS application that converts raw DNA data from Ancestry.com into raymarching shader art, transformed into Ethereum-based NFTs. Implemented user login system deployed on Firebase.



SHADER PARK

SEPT 2019 - FEB 2020

Resolved front-end issues in Vue.js; collaborated with stakeholders to design and implement frontend modules. Utilized GLSL for shader development.



CARD SWAPPER

SEPT 2017 - DEC 2019

Built Objective-C mobile app for instant gift card trading.

AWARDS



RAYLS HACKATHON

NOV 18-19, 2025, BUENOS AIRES, ARGENTINA

PRAXOS | 1ST PLACE WINNER

- Won 1st place for building AI-generated ERC-4626 vault engine for RWAs
- Designed compliant vault system integrating ERC-3643 + identity modules
- Built institutional RWA tokenization + allocation optimization engine



ETHGLOBAL CANNES

JUL 4-6, 2025, CANNES, FRANCE

LIVESTAKES | FINALIST

- Built real-time prediction market for live hackathon demos
- Implemented AI-driven market creation from livestream data
- Enabled instant payouts via Flow smart contracts



ETHGLOBAL BUENOS AIRES

NOV 21-23, 2025, BUENOS AIRES, ARGENTINA

HEDGEPOD | WORLD POOL PRIZE WINNER

- Built cross-chain AI DeFi agent managing yield across 8+ chains
- Implemented APR-aware routing + automated rebalancing logic
- Delivered gasless consumer UX for 23M+ World App users

EDUCATION



UNIVERSITY OF WASHINGTON

SEP 2014 - AUG 2017, SEATTLE, WA

BACHELORS OF SCIENCE

Molecular Cellular Developmental Biology (MCDB)
Computer Science Software Engineering Minor (CSSE)



BRAINSTATION

JAN 2020 - APR 2020, ONTARIO, CA

WEB DEVELOPMENT DIPLOMA

React, Javascript, SCSS, HTML5, SQL, Front End Responsive Design.
Worked with PMs, UX designers, Data Science, Digital Marketing Team on Projects.

- > Playwright
- > Pytest
- > Hardhat
- > Foundry
- > Property-Based
- > Invariant
- > Concurrency
- > Snapshot Regression Testing

Program Management & Agile

- > Jira
- > Scrum
- > Kanban
- > Linear
- > MS SQL Server
- > PostgreSQL

Game Development & Creative Coding

- > GLSL
- > Three.js
- > Shaderpark
- > Unity
- > Unreal Engine
- > Canvas API